DOM207-DOM3007-DOM6401: MP4

BACKGROUND

About 12 years back, when I was working as a scientist at the Department of Science and Technology (DST), Government of India, I was engaged in a government project that concerned Indian women's participation in the job market. It was an interesting project, on which I had four other scientists working with me. The data collection team was another government agency external to my team.

Your MP4 is a much, much simplified version of that project.

INTRODUCTION:

- ✓ In this project, imagine that you're members of the project team in a DSA consulting firm named DOM207-DOM3007-DOM6401, working on a project contracted out by your client – the Government of India.
- ✓ Imagine that the data set ***IWomenJob.xlsx*** has been collected for you by a 3rd party data collection agency appointed by the Government of India. **This is proprietary data, so you're NOT allowed to add any new data to this file**

WORK AND SUBMISSION REQUIREMENTS AND GUIDELINES

- ✓ For this project, you must work in a group of **no more than 2 (only in an exceptional case 3) students per group**.
- ✓ Solo work is also allowed and appreciated.
- ✓ Each group must submit only one project report **on BB**.
- ✓ Each submission must contain the following items:
    1) A report in word doc or PDF with an appropriate title page. This page should contain a title for the project, **your name and your collaborator name(s)**, your roll #, course name and #, and your instructor's name
    2) The report must be divided into at least the following sections (**section heading MANDATORY**):
    3) **Introduction** (say what you're set out to achieve in this project)
    4) **Data preprocessing** (provide details of all data cleansing steps you've undertaken, if any)
    5) **Data visualization** (provide some worthwhile charts, diagrams, etc. for visualization – do not provide useless diagrams – you'll be penalized if you do)
    6) **Hypotheses or theory building** (if appropriate)
    7) **Analysis** (Verification of model assumptions, and **complete** technical analysis with justifications for all of your findings). You may insert a few helpful screenshots of your work (C/P from your output). These must be relevant and persuasive. **Do not put rubbish diagrams or figures – you'll be penalized if you do!!!**
    8) **Recommendations** **Your final recommendations with proper justifications** (based on your findings)
    9) **References**
    10) **Submit the project report (Word doc or PDF) and your supporting R or Python files containing the complete analysis (project files and .R or .py files)**.

11) There's no fixed requirement for word or page limit for the report. However, the material in it should be substantial and meaningful, exactly like a project report you'd submit to your client
    - **NOTE**: **SUBMISSION ON BB is completely your responsibility**. **Submit well BEFORE the deadline. Last-minute technical failures (BB problems; network outages; computer crashes; dogs or cats or grandpas or grandmas or interviews) or sickness will NOT be considered in any way (if you're sick, ask your partner ASAP to submit). Failure to conform to these requirements results in a zero, meaning in the corporate context, your company loses the contract. So, be responsible and work in a professional capacity.**

VERY IMPORTANT:

- **Remember, DOM207-DOM3007-DOM6401 is a skill-development course that will help you in good placements in this field**
- So, if you do not follow these directions, **you'll be heavily penalized**. The project must be as formal as possible, as is common in the industry
- **The grading will be at a professional level**
- **Project submission deadline is inflexible and strictly enforced!!**

PROJECT FOR DOM207-DOM3007-DOM6401

This project is about Indian women's participation in the job market. The women in the sample have several characteristics. The question that you should be concerned with in this project is, what factors constitute the primary determinants of women's participation in the job market.

As I said before in class, the project is not easy – you have to consider lots of things:

- Hypotheses should be driven by theory and logical arguments.
- Do **NOT** do ***harking***. If you're caught, you won't get any credit for MP4.
- You need to check all sorts of things like data consistency, descriptive stats, validity of assumptions, MC, HS, interactions, quadratics, and so on and so forth.
- Dataviz is also important
- Create a new variable called $\boldsymbol{nwifeinc} = \frac{(\boldsymbol{faminc} - \boldsymbol{sal} * \boldsymbol{hours})}{\boldsymbol{1000}}$ and include it in your model. What is its role in the model and what are its implications for the results?
- Your discussions, conclusions, and recommendations must be entirely based on your findings – not idle speculation!!!

The actual data file is ***IWomenJob.xlsx***.